NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

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TOPIC: Project Estimation Techniques cont…



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CONCEPTS COVERED

- Some different types of estimation techniques



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Types of Estimation Techniques

- Though there are many techniques of estimating, they can broadly be classified into:
  - Top-down
  - Bottom-up
- What about:
  - **Algorithmic models?**
  - **Expert opinion ?**
  - **Analogy ?**
  - **Price to win?**





Bottom-up versus top-down

- Bottom-up
  - identify all tasks that have to be done – so quite time-consuming
  - use when you have no data about similar past projects
- Top-down
  - produce overall estimate based on project cost drivers based on past project data
  - divide overall estimate between jobs to be done





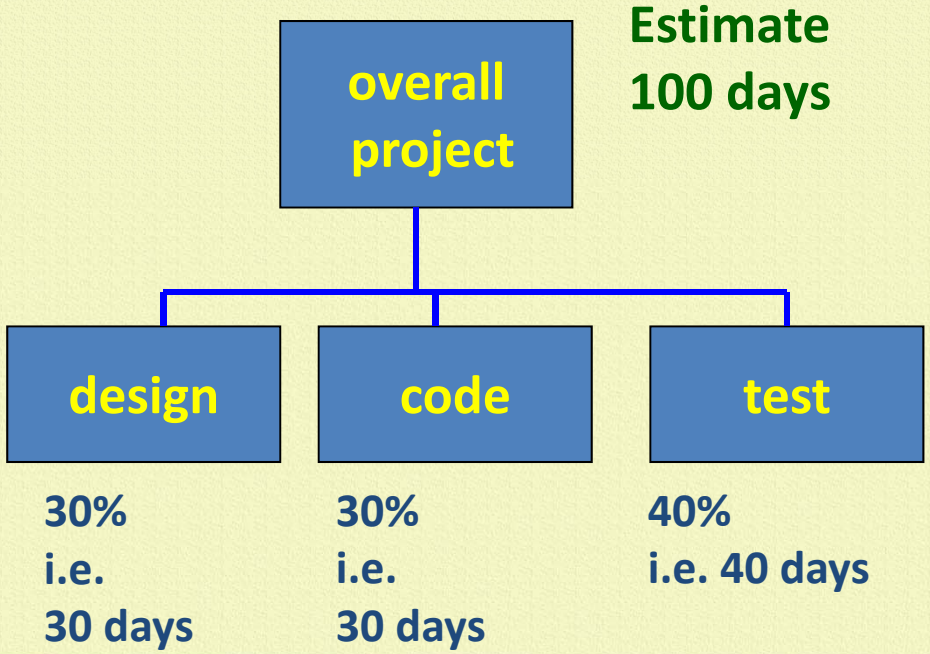
Bottom-up estimating

1. Break the project activities into smaller and smaller components
   - Stop when you get to what one person can do in one/two weeks
2. Estimate costs for the lowest level activities
3. At each higher level calculate estimate by adding estimates for lower levels



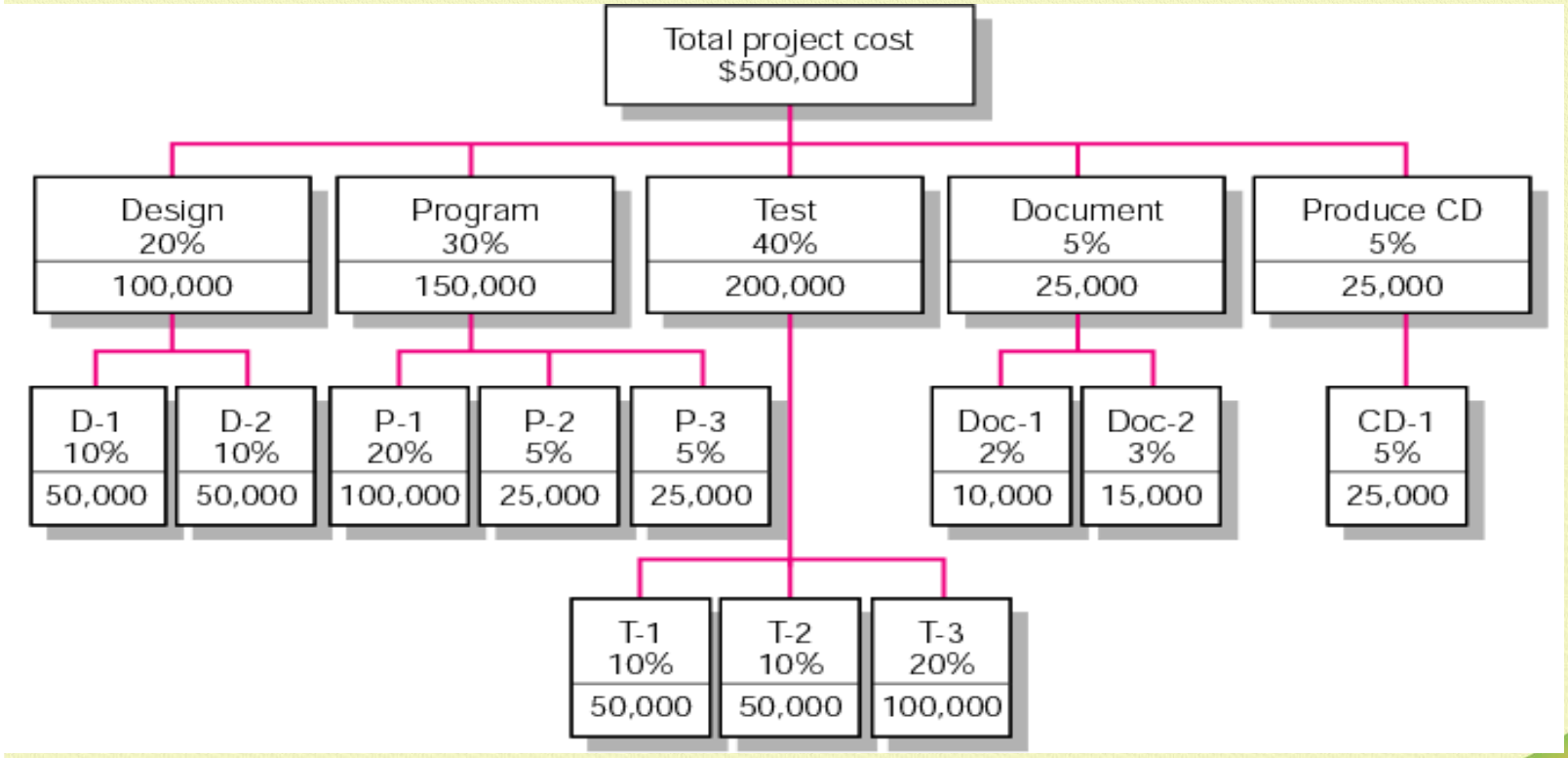


Top-down estimates



- Produce overall estimate using effort driver(s)
- Distribute proportions of overall estimate to components

Top-down Example







Top-Down Estimating: Pros

- It accounts for system-level activities such as integration, documentation, configuration management, etc.:
  - **Many of these may be ignored in bottom-up estimating methods.**
- It requires minimal project details:
  - It is usually faster, easier to implement.





Top-Down Estimating: Cons

- It often does not take into consideration the difficult low-level problems:
  - **Tends to underestimate and overlook complexities of low-level components.**
- It provides no detailed basis for justifying decisions or estimates.





Bottom-up Estimating: Pro

- It permits the software group to estimate in an almost traditional fashion:
  - Each group estimates components for which it has adequate experience.
- It is more stable because the estimation errors in the various components more or less balance out.





Bottom-up Estimating: Con

- It may overlook many of the system-level costs:
  - Integration,
  - configuration management,
  - quality assurance, etc.
- It may be inaccurate because the necessary information may not be available in the early phase.
- It tends to be more time-consuming.





Criteria for a Good Algorithmic Model

- Defined—clear procedure
- Accurate
- Objective—avoids subjective factors
- Results understandable
- Stable— Results valid for a wide range of parameter values
- Easy to Use
- Causal—future data not required





Algorithmic Models

- For project planning, we need:
  - Effort (cost)
  - Duration
- Hard to estimate effort (or cost) or duration directly from a problem description.
- Effort and Duration can be measured in terms of certain project characteristics that correlate with it:
  - Called size





Software Size

- What exactly is the size of a software project?
  - How do you measure it?
- Any characteristic of software that is easily measured and correlates with effort.
  - SLOC
  - Function point



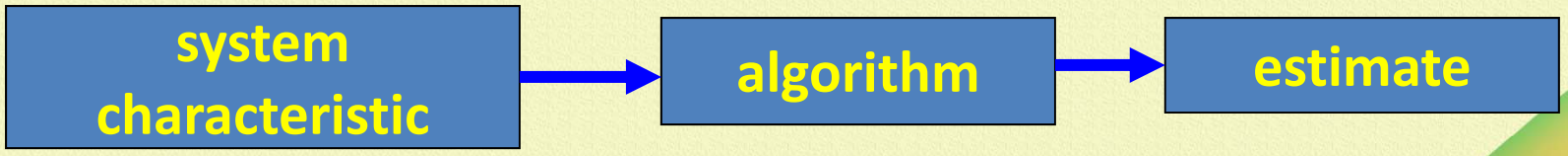


Algorithmic/Parametric models

- COCOMO (lines of code) and function points examples of these
- A Problem with LOC based models (COCOMO etc):



but what is desired is







Algorithmic Methods: Pro

- It is able to generate repeatable estimations.
- It is easy to modify input data, refine and customise formulas.
- It is efficient and able to support a family of estimations or a sensitivity analysis.
- It can be meaningfully calibrated to previous experience.





Algorithmic Methods: Cons

- It lacks means to deal with exceptional conditions, such as:
  - exceptional teamwork, exceptional match between skill-levels and tasks, etc.
- Poor sizing inputs and inaccurate cost driver rating will result in inaccurate estimation.
- Some factors such as experience cannot be easily quantified.





Model Calibration

- Many models are developed for specific situations and are, by definition, calibrated to that situation.
- Such models usually are not useful outside of their particular environment.
  - Calibration is needed to increase the accuracy of one of these general models.
  - Calibration is in a sense customising a generic model.
- Items that can be calibrated in a model include:
  - product types, operating environments, labour rates and factors, various relationships between functional cost items, etc.

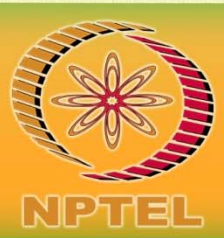




Some recommendations

- Do not depend on a single cost or schedule estimate.
- Use several estimating techniques or cost models:
  - Compare the results and determine the reasons for any large variations.
- Document the assumptions made when making the estimates.
- Monitor the project to detect when assumptions that turn out to be wrong jeopardize the accuracy of the estimate.
- Improve software process:
  - **Maintain a historical database**





Simplistic Model

- **estimated effort = (system size) / productivity**
- As example:

  system size = lines of code

  productivity = lines of code per day

What is wrong with the simplistic model?

- **productivity = (system size) / effort**
  - based on past projects





Example 1

Consider a transaction project of 38,000 lines of code, what is the shortest time it will take to develop? Productivity is about 400 SLOC/staff month

Effort = $(\text{productivity})^{-1}$ (size)

= (1/.400 KSLOC/SM) (38 KSLOC)

= 2.5 (38) ≈ 100 SM

Min time = .75 T= $(.75)(2.5)(\text{SM})^{1/3}$

≈ $1.875(100)^{1/3}$

≈ 1.875 x 4.63 ≈ 9 months

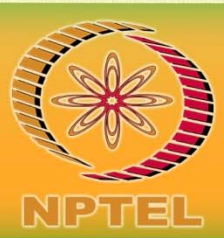




Summary

- Discussed some different types of estimation techniques such as top-down and bottom-up estimation techniques
- Also discussed the algorithmic / parametric models for project estimation.





References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.
2. R. Mall, *Fundamentals of Software Engineering*, Fifth Edition, PHI Learning Pvt. Ltd., 2018.






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